

# J Dhana Santhosh Reddy

## AI/ML Engineer

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### SUMMARY

**AI/ML Engineer with around 4 years of experience** designing, developing, and deploying scalable machine learning, deep learning, and Generative AI solutions across financial and enterprise domains. Skilled in building and fine-tuning LLMs and RAG pipelines using LangChain, Hugging Face Transformers, and vector databases such as FAISS, Pinecone, and Chroma. Proficient in Python, scikit-learn, XGBoost, LightGBM, and SHAP for predictive modeling, time-series forecasting, and explainable AI. Experienced in architecting end-to-end ETL pipelines using Apache Airflow, SQL, and AWS services to enable real-time analytics. Adept at deploying AI models through Docker, Kubernetes, and CI/CD pipelines within AWS and Azure environments. Strong collaborator skilled at aligning AI initiatives with business goals to enhance forecasting accuracy, automation, and decision-making efficiency.

### WORK EXPERIENCE

#### Morgan Stanley, USA | AI/ML Engineer

Nov 2024 - Current

- Designed and deployed Generative AI and Large Language Model (LLM) solutions using LangChain, Hugging Face Transformers, and PyTorch, enabling automation of financial document analysis and client communication workflows.
- Developed RAG pipelines integrating vector databases to enhance context retrieval for regulatory, compliance, and investment applications.
- Built predictive and classification models using LightGBM, XGBoost, and Logistic Regression to identify trading anomalies, customer churn patterns, and cross-sell opportunities, improving model accuracy and interpretability.
- Designed and implemented time-series forecasting frameworks using Azure Machine Learning, AutoML, and XGBoost Regressor to predict revenue volatility, liquidity ratios, and credit risk exposure, enabling proactive financial planning and capital allocation.
- Developed serverless ETL workflows leveraging Azure Data Factory, Azure Synapse Analytics, and Data Lake Storage Gen2 to orchestrate ingestion, transformation, and validation of high-volume financial transaction and market feed data.
- Deployed and monitored containerized ML pipelines on Azure Kubernetes Service (AKS) with integrated Azure DevOps CI/CD for automated retraining, versioning, and real-time model scoring.
- Integrated Azure Monitor, Log Analytics, and Application Insights for end-to-end observability and model performance tracking across distributed environments.
- Integrated speech recognition, NLP, and computer vision modules into multi-modal AI assistants for internal analytics, enhancing productivity and accessibility across research teams.
- Collaborated with risk, data, and compliance teams to enforce model governance, explainability (XAI), and security frameworks, ensuring adherence to HIPAA, SOC 2, and internal audit standards.

#### Rlogical Techsoft Pvt. Ltd, India | AI/ML Engineer

Jan 2021 - July 2023

- Designed, developed, and deployed machine learning classification models using Python, scikit-learn, and XGBoost to predict customer churn and drive retention strategies through actionable business insights.
- Engineered robust feature engineering pipelines with Pandas and NumPy, leveraging encoding, binning, and dimensionality reduction (PCA) techniques to optimize data quality and improve model performance.
- Automated ETL workflows and data ingestion pipelines using Apache Airflow, SQL, and AWS S3, ensuring high data integrity, reduced latency, and consistent pipeline reliability for analytics applications.
- Enhanced model explainability through SHAP and LIME frameworks, translating complex ML outputs into interpretable insights for stakeholders and supporting data-driven decision-making.
- Optimized model accuracy and generalization using hyperparameter tuning frameworks like Optuna and GridSearchCV, implementing k-fold validation to prevent overfitting and ensure robust predictions.
- Deployed scalable ML solutions using Docker containers and AWS SageMaker, integrating Flask/FastAPI for real-time inference APIs and automating CI/CD processes through Jenkins and GitHub Actions.
- Implemented MLOps best practices by integrating MLflow for experiment tracking, model registry, and lifecycle management, streamlining collaboration between data science and DevOps teams.
- Collaborated cross-functionally with product managers and data engineers to align ML model outputs with CRM, marketing, and financial objectives, improving personalization and targeting efficiency.
- Applied time-series forecasting models (ARIMA, Prophet, XGBoost Regressor) for revenue trend prediction and financial risk analysis, supporting business planning and operational optimization.
- Ensured data privacy and compliance with GDPR and HIPAA standards by anonymizing PII and implementing secure cloud-based data handling practices in production ML environments.

### TECHNICAL SKILLS

- Methodologies / Languages:** SDLC, Agile, Waterfall, Python, C++, Scala, GoLang.
- AI/ML Domain:** Natural Language Processing, Computer Vision, Image Processing, Deep Learning, Machine Learning, Reinforcement Learning, Expert Systems.
- AI/ML Technologies:** Large language models (LLMs), Retrieval Augmented Generation (RAG), Generative AI, Pytorch, TensorFlow, Keras, OpenNN, TorchVision, OpenCV, NLTK, SpaCy, Hugging Face, LangChain, Model Interpretability, Explainable AI, scikit-learn, Tensorboard, OpenAI Gym, Pandas, Matplotlib, Statistical Analysis, ChatGPT, BERT, Fine Tuning, Prompt Engineering, Conversational AI, Ranking, query classification, Named Entity Recognition (NER), Gradient Boosting Trees, PCA, Information Retrieval, Seq2Seq, Knowledge Graph, Linear Regression, Logistic Regression, Random Forest, Association Rules, Support Vector Machine, ggplot2, Pandas.
- Software Technologies:** SQL, MySQL, MongoDB, RESTful API, Scala Play, FastAPI, Swagger, Flask, AWS, Sage Maker, ElasticSearch, Agile, CI/CD, Design Patterns, Concurrency, Git, GitLab, Linux, Windows, A/B Testing.
- Operating System:** Windows, Linux

### EDUCATION

**Master's in Robotics** | University of Maryland, College Park, MD

**Bachelor's in Mechatronics** | SRM Institute of Science and Technology, Chennai, Tamil Nadu